

Setting Up

The Droste Maker program has only one function, `makeDroste()`, which accepts a `Pillow Image` object or a string of an image's filename. The function returns a `Pillow Image` object with any magenta pixels recursively replaced by a version of the same image:

Python

```
from PIL import Image

def makeDroste(baseImage, stopAfter=10):
    # If baseImage is a string of an image filename, load that image:
    if isinstance(baseImage, str):
        baseImage = Image.open(baseImage)
```

The program begins by importing the `Image` class from the `Pillow` library (named `PIL` as a Python module). Within the `makeDroste()` function, we check whether the `baseImage` parameter is a string, and if so, we replace it with a `Pillow Image` object loaded from the corresponding image file.

Next, we check whether the `stopAfter` parameter is `0`. If it is, we've reached one of the algorithm's base cases and the function returns the `Pillow Image` object of the base image:

Python

```
if stopAfter == 0:
    # BASE CASE
    return baseImage
```

The `stopAfter` parameter is `10` by default if the function call doesn't provide one. The recursive call to `makeDroste()` later in this function passes `stopAfter - 1` as the argument for this parameter so that it decreases with each recursive call and approaches the base case of `0`.

For example, passing `0` for `stopAfter` results in the function immediately returning a recursive image identical to the base image. Passing `1` for